

MATHEMATICS-III

Unit-3: Number Theory & Cryptography

Comprehensive Study Notes

Unit-3: Number Theory & Cryptography

1. Primes

1.1 Prime Numbers

A **prime number** is a natural number greater than 1 that has no positive divisors other than **1 and itself**.

Examples: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, ...

Composite Numbers: Natural numbers greater than 1 that are not prime.

Examples: 4, 6, 8, 9, 10, 12, 14, 15, ...

Note: 1 is neither prime nor composite.

1.2 Properties of Prime Numbers

Property	Description
Unique Factorization	Every integer > 1 can be uniquely expressed as a product of primes
Infinite Primes	There are infinitely many prime numbers (Euclid's proof)
Smallest Prime	2 is the smallest prime and the only even prime

1.3 Sieve of Eratosthenes

An ancient algorithm for finding all primes up to a given limit.

Algorithm:

1. Write all numbers from 2 to n .
2. Start with the smallest prime, 2.
3. Mark all multiples of 2 (except 2) as composite.
4. Move to the next unmarked number (3) and mark all its multiples.

5. Continue until you reach the square root of n .
6. All unmarked numbers are prime.

Example: Find primes up to 30.

Numbers: 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30
Step 1: Mark multiples of 2: 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30
Step 2: Mark multiples of 3: 9, 15, 21, 27
Step 3: Mark multiples of 5: 25
Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29

2. Primality Testing

2.1 Trial Division Method

Test divisibility by all integers from 2 to $\sqrt{n}$.

Algorithm:

```
If n < 2: return false
For i = 2 to sqrt(n):
    If n % i == 0: return false
Return true
```

Time Complexity: $O(\sqrt{n})$

2.2 Fermat Primality Test

Based on **Fermat's Little Theorem**:

If p is prime and a is not divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$

Algorithm:

```
For a random a:
    Compute  $a^{n-1} \bmod n$ 
    If result  $\neq 1$ : n is composite
    Else: n may be prime (with some probability)
```

Limitations: Carmichael numbers (561, 1105, 1729, ...) pass the test for all a , despite being composite.

2.3 Miller-Rabin Primality Test

A more reliable probabilistic primality test.

Algorithm:

1. Write $n-1 = 2^s * d$ where d is odd
2. Choose random a such that $1 < a < n-1$
3. Compute $x = a^d \bmod n$
4. If $x = 1$ or $x = n-1$: continue to next a
5. Repeat $s-1$ times:
 $x = x^2 \bmod n$
 If $x = n-1$: continue to next a
6. Return "composite"
7. If all a pass: n is probably prime

3. Factorization

3.1 Trial Division

Divide by all primes up to $\sqrt{n}$.

Example: Factor 84

```
84 / 2 = 42
42 / 2 = 21
21 / 3 = 7
7 / 7 = 1
84 = 2^2 x 3 x 7
```

3.2 Pollard's Rho Algorithm

A randomized algorithm for integer factorization.

Algorithm:

```
1. Choose  $f(x) = x^2 + 1 \pmod{n}$ 
2. Set  $a = 2, b = 2$ 
3. Repeat:
     $a = f(a)$ 
     $b = f(f(b))$ 
     $d = \gcd(|a-b|, n)$ 
    If  $d > 1$  and  $d < n$ : return  $d$  (a factor)
    If  $d == n$ : restart with different function
```

3.3 Comparison of Factorization Methods

Method	Best For	Complexity
Trial Division	Small numbers	$O(\sqrt{n})$
Pollard's Rho	Medium numbers	$O(n^{1/4})$
Quadratic Sieve	Large numbers	Subexponential

Number Field Sieve	Very large numbers	Subexponential
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4. Chinese Remainder Theorem

4.1 Statement

Let $m_1, m_2, \dots, m_k$ be pairwise coprime integers. Then the system of congruences:

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\x &\equiv a_2 \pmod{m_2} \\&\vdots \\x &\equiv a_k \pmod{m_k}\end{aligned}$$

has a unique solution modulo $M = m_1 \cdot m_2 \cdot \dots \cdot m_k$.

4.2 Solution Algorithm

Step 1: Compute $M = m_1 \cdot m_2 \cdot \dots \cdot m_k$

Step 2: For each i :

- Compute $M_i = M / m_i$
- Compute $y_i = M_i^{-1} \pmod{m_i}$

Step 3: Compute:

$$x = \sum_{i=1}^k a_i \cdot M_i \cdot y_i \pmod{M}$$

4.3 Solved Examples

Example 1: Solve:

$$\begin{aligned}x &\equiv 2 \pmod{3} \\x &\equiv 3 \pmod{5} \\x &\equiv 2 \pmod{7}\end{aligned}$$

Solution:

Step 1: $M = 3 \times 5 \times 7 = 105$

Step 2:

- $M_1 = 105/3 = 35, y_1 = 35^{-1} \pmod{3}$
 $35 \pmod{3} = 2$, inverse of 2 mod 3 is 2 $\Rightarrow y_1 = 2$
- $M_2 = 105/5 = 21, y_2 = 21^{-1} \pmod{5}$
 $21 \pmod{5} = 1$, inverse of 1 mod 5 is 1 $\Rightarrow y_2 = 1$
- $M_3 = 105/7 = 15, y_3 = 15^{-1} \pmod{7}$
 $15 \pmod{7} = 1$, inverse of 1 mod 7 is 1 $\Rightarrow y_3 = 1$

Step 3:

$$\begin{aligned} x &= 2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1 \\ &= 140 + 63 + 30 = 233 \\ &= 233 \pmod{105} = 23 \end{aligned}$$

Answer: $x \equiv 23 \pmod{105}$

Example 2: Solve:

$$\begin{aligned} x &\equiv 1 \pmod{2} \\ x &\equiv 2 \pmod{3} \\ x &\equiv 3 \pmod{5} \end{aligned}$$

Solution:

Step 1: $M = 2 \times 3 \times 5 = 30$

Step 2:

- $M_1 = 30/2 = 15, y_1 = 15^{-1} \pmod{2} \Rightarrow 15 \pmod{2} = 1$, inverse of 1 is 1 $\Rightarrow y_1 = 1$
- $M_2 = 30/3 = 10, y_2 = 10^{-1} \pmod{3} \Rightarrow 10 \pmod{3} = 1$, inverse of 1 is 1 $\Rightarrow y_2 = 1$
- $M_3 = 30/5 = 6, y_3 = 6^{-1} \pmod{5} \Rightarrow 6 \pmod{5} = 1$, inverse of 1 is 1 $\Rightarrow y_3 = 1$

Step 3:

$$\begin{aligned} x &= 1 \times 15 \times 1 + 2 \times 10 \times 1 + 3 \times 6 \times 1 \\ &= 15 + 20 + 18 = 53 \end{aligned}$$

$$= 53 \bmod 30 = 23$$

Answer: $x \equiv 23 \pmod{30}$

5. Quadratic Congruence

5.1 Definition

A **quadratic congruence** is a congruence of the form:

$$ax^2 + bx + c \equiv 0 \pmod{n}$$

The most common form is $x^2 \equiv a \pmod{p}$ where p is prime.

5.2 Quadratic Residues

A number a is a **quadratic residue** modulo n if there exists some x such that:

$$x^2 \equiv a \pmod{n}$$

If no such x exists, a is a **quadratic non-residue**.

5.3 Legendre Symbol

The Legendre symbol (a/p) is defined for odd prime p :

$$\begin{aligned} &1 \text{ if } a \text{ is a quadratic residue mod } p \\ &-1 \text{ if } a \text{ is a quadratic non-residue mod } p \\ &(a/p) = 0 \text{ if } p \text{ divides } a \end{aligned}$$

5.4 Properties of Legendre Symbol

Property	Formula
Multiplicative	$(ab/p) = (a/p) \cdot (b/p)$
Euler's Criterion	$(a/p) \equiv a^{(p-1)/2} \pmod{p}$
For -1	$(-1/p) = (-1)^{(p-1)/2}$

For 2

$$(2/p) = (-1)^{(p^2-1)/8}$$

5.5 Euler's Criterion

$$(a/p) \equiv a^{(p-1)/2} \pmod{p}$$

Example: Determine if 2 is a quadratic residue modulo 7.

$$2^{(7-1)/2} = 2^3 = 8 \equiv 1 \pmod{7}$$

Since result is 1, 2 is a quadratic residue modulo 7. Check: $3^2 = 9 \equiv 2 \pmod{7}$ ✓

6. Modular Exponentiation

6.1 Modular Exponentiation

Modular exponentiation involves computing $a^b \bmod n$.

Naive Method:

```
result = 1
for i = 1 to b:
    result = (result * a) % n
```

Time Complexity: $O(b)$ (Inefficient for large b)

6.2 Fast Exponentiation (Binary Exponentiation)

Algorithm:

```
function pow_mod(a, b, n):
    result = 1
    while b > 0:
        if b % 2 == 1:
            result = (result * a) % n
        a = (a * a) % n
        b = b // 2
    return result
```

Time Complexity: $O(\log b)$

Example: Compute $3^{13} \bmod 5$

```
b = 13 (binary: 1101)
b is odd: result = 1 x 3 = 3
a = 3^2 = 9 ≡ 4, b = 6
b is even: a = 4^2 = 16 ≡ 1, b = 3
b is odd: result = 3 x 1 = 3
a = 1^2 = 1, b = 1
b is odd: result = 3 x 1 = 3
```

$$a = 1^2 = 1, b = 0$$

$$\text{Result: } 3^{13} \equiv 3 \pmod{5}$$

Check: $3^{13} = 1594323, 1594323 \bmod 5 = 3 \checkmark$

6.3 Applications of Modular Exponentiation

Application	Description
RSA Encryption	Encryption and decryption operations
Digital Signatures	Signature generation and verification
Primality Testing	Fermat and Miller-Rabin tests
Diffie-Hellman	Key exchange protocol

7. HPU Important Questions

Short Answer Questions (2–5 Marks)

1. Define a prime number.
2. What is the Sieve of Eratosthenes?
3. State Fermat's Little Theorem.
4. State the Chinese Remainder Theorem.
5. What is a quadratic residue?
6. Define the Legendre symbol.
7. What is Euler's Criterion?
8. What is modular exponentiation?
9. What is the complexity of binary exponentiation?
10. What is Pollard's Rho algorithm?

Long Answer Questions (10 Marks)

1. Explain the Sieve of Eratosthenes with an example.
2. State and prove the Chinese Remainder Theorem.
3. Explain Fermat's Little Theorem and its applications.
4. Explain modular exponentiation using the fast exponentiation method.
5. Explain Pollard's Rho factorization algorithm.
6. Solve the system of congruences: $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$.
7. Determine if 3 is a quadratic residue modulo 7.

Objective Questions (1 Mark)

1. Which of the following is a prime number?

(a) 1 (b) 2 (c) 4 (d) 9

Answer: (b)

2. The Sieve of Eratosthenes is used to find:

(a) GCD (b) LCM (c) Prime numbers (d) Factorials

Answer: (c)

3. Fermat's Little Theorem states:

(a) $a^p \equiv 1 \pmod{p}$ (b) $a^{p-1} \equiv 1 \pmod{p}$ (c) $a^{p+1} \equiv 1 \pmod{p}$ (d) None

Answer: (b)

4. The Chinese Remainder Theorem solves:

- (a) Linear congruences (b) Quadratic congruences (c) Systems of congruences
(d) None

Answer: (c)

5. The Legendre symbol $(2/p)$ equals:

- (a) 1 (b) -1 (c) $(-1)^{(p^2-1)/8}$ (d) 0

Answer: (c)

Quick Revision

Key Algorithms Summary

Algorithm	Purpose	Complexity
Sieve of Eratosthenes	Find primes up to n	$O(n \log \log n)$
Trial Division	Primality testing	$O(\sqrt{n})$
Pollard's Rho	Factorization	$O(n^{1/4})$
Binary Exponentiation	$a^b \bmod n$	$O(\log b)$

Key Theorems

Theorem	Formula
Fermat's Little Theorem	$a^{p-1} \equiv 1 \pmod{p}$
Euler's Criterion	$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$
Chinese Remainder	$x = \sum a_i M_i y_i \pmod{M}$

CRT Formula

$$x = \sum_{i=1}^k a_i \cdot M_i \cdot y_i \pmod{M}$$

where $M_i = M/m_i$ and $y_i = M_i^{-1} \pmod{m_i}$

End of Unit-III Notes — MATHEMATICS-III (HPU BCA)